An exactly solvable case of $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$: a Pöschl–Teller well in a box

Nadav Jean · summary for the supervisor meeting, 28.10.2026 · derivations in `PT_Box_Derivation.pdf`

1 The system

A particle of mass m in a Pöschl–Teller well placed in the middle of an infinite square well of length L:

$$V(x) = -\frac{\hbar^2}{2m}\alpha^2\frac{\lambda(\lambda-1)}{\cosh^2(\alpha x)} \xrightarrow{\lambda=2} -\frac{\hbar^2\alpha^2}{m}\frac{1}{\cosh^2(\alpha x)} \quad (|x|<L/2), \qquad V=+\infty \quad (|x|\geq L/2). \tag{1}$$

The well is the "modified Pöschl–Teller well" written exactly as in Table I of Deffner, Jarzynski and del Campo [1], who take it from Flügge [2]. It has depth $\hbar^2\alpha^2/m$ and width $\sim 1/\alpha$. We use $\lambda = 2$, the simplest *reflectionless* case: a wave crosses the well without any reflection [3, 4]. That single property is what makes every result below exact. Only two numbers matter: αL (box length in well widths) and $k_BT/(\hbar^2\alpha^2/2m)$.

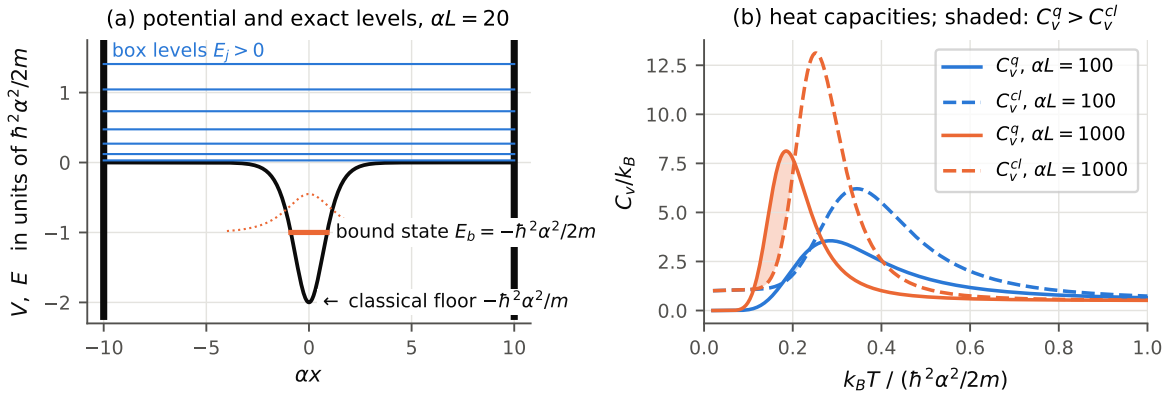


Figure 1: (a) The potential (black), the box levels (blue) and the single bound state (orange, with its wavefunction $\propto 1/\cosh\alpha x$ dotted), for $\alpha L = 20$. (b) C_v^{q} (solid) and C_v^{cl} (dashed) for $\alpha L = 100$ and 1000; shaded where $C_v^{\mathrm{q}} > C_v^{\mathrm{cl}}$.

2 Energy levels (exact)

- **One bound state, halfway up the well** [3]: $E_b = -\hbar^2\alpha^2/2m$, with wavefunction $\propto 1/\cosh(\alpha x)$. The other half of the depth is zero-point energy.
- **Box levels** [4, 5]: $E_j = \hbar^2 k_j^2/2m$, $j = 1, 2, \ldots$, with

$$k_jL = j\pi + 2\arctan\left(\frac{k_j}{\alpha}\right). \tag{2}$$

Both hold up to corrections of order $e^{-\alpha L}$ (the exact finite-L equations are in the derivation). *Heuristic:* without the well, $k_jL = j\pi$. Far from the well, all the well does to a wave is shift its phase by $2\arctan(k/\alpha)$, which pushes every box level up by less than one spacing. As $k\to\infty$ the shift tends to π, so level j moves up to where level $j+1$ of the empty box was: exactly one state has left the continuum, and it is the bound state (Levinson's theorem [6]).

3 Quantum heat capacity

$Z_q = e^{-\beta E_b} + \sum_j e^{-\beta E_j}$ and $C_v^{\mathrm{q}} = k_B\beta^2\,\mathrm{Var}(E)$ [7]. Writing the sum over j as an integral over k (Poisson summation [8]), with the level density $\mathrm{d}j/\mathrm{d}k$ read off Eq. (2) (the Beth–Uhlenbeck idea [9]), every integral is elementary [10]:

$$Z_q = \frac{L}{\lambda_{\mathrm{th}}} - \frac{1}{2} + e^{a/2}\,\mathrm{erf}\,\sqrt{a/2}, \qquad a \equiv \frac{\beta\hbar^2\alpha^2}{m} = \frac{\text{depth}}{k_BT}, \qquad \lambda_{\mathrm{th}} = \sqrt{\frac{2\pi\hbar^2\beta}{m}}, \tag{3}$$

$$\frac{C_v^{\mathrm{q}}}{k_B} = a^2\frac{\partial^2\ln Z_q}{\partial a^2} = a^2\left[\frac{Z_q''}{Z_q} - \left(\frac{Z_q'}{Z_q}\right)^2\right], \quad Z_q' = -\tfrac{1}{2a}\tfrac{L}{\lambda_{\mathrm{th}}} + G', \quad G = e^{a/2}\,\mathrm{erf}\,\sqrt{a/2}, \quad G' = \tfrac{1}{2}G + (2\pi a)^{-1/2},$$
$$Z_q'' = \tfrac{3}{4a^2}\tfrac{L}{\lambda_{\mathrm{th}}} + G'', \qquad G'' = \tfrac{1}{2}G' - \tfrac{1}{2}(2\pi)^{-1/2}a^{-3/2}. \tag{4}$$

Reading of Eq. (3)*:* L/λ_{th} is a free particle in the box; $-\frac{1}{2}$ is the hard-wall correction; $e^{a/2}$ is the bound state; and $-e^{a/2}(1-\mathrm{erf})$ removes the one continuum state that became the bound state, spread over energies by the phase shift. Eq. (4) agrees with $\mathrm{Var}(E)$ summed over the exact levels to $2\times10^{-13}k_B$.

4 Classical heat capacity

$Z_{\rm cl} = \frac{1}{2\pi\hbar}\int\mathrm{d}x\int\mathrm{d}p\; e^{-\beta H} = \frac{1}{\lambda_{\rm th}}\int e^{-\beta V}\mathrm{d}x$ [7]. Splitting off the flat part of the box and substituting $t=\tanh(\alpha x)$:

$$Z_{\rm cl} = \frac{L+I}{\lambda_{\rm th}}, \qquad \alpha I = \int_{-1}^{1}\frac{e^{a(1-t^2)}-1}{1-t^2}\,\mathrm{d}t = \sum_{\ell\ge 1}\frac{(2a)^\ell}{\ell\,(2\ell-1)!!}, \tag{5}$$

$$\frac{C_v^{\rm cl}}{k_B} = \frac{1}{2} + a^2\left[\frac{\alpha J_2}{\alpha L+\alpha I} - \left(\frac{\alpha J_1}{\alpha L+\alpha I}\right)^2\right], \qquad \alpha J_1 = \sqrt{\pi}\,\frac{e^a\,\mathrm{erf}\sqrt{a}}{\sqrt{a}}, \qquad \alpha J_2 = \alpha J_1\left(1-\frac{1}{2a}\right)+\frac{1}{a}. \tag{6}$$

Reading: $\frac{1}{2}k_B$ is the kinetic (equipartition) part, and the bracket is $\beta^2\mathrm{Var}(V)$, the potential-energy fluctuations. At low T the particle sits in the harmonic bottom of the well and $C_v^{\rm cl}\to k_B$; at high T the well is irrelevant and $C_v^{\rm cl}\to k_B/2$. Eq. (6) agrees with direct quadrature of the phase-space integral to $10^{-9}k_B$.

5 Why $C_v^{\rm q} > C_v^{\rm cl}$: a heuristic

Each curve in Fig. 1(b) has one peak: *desorption*, the particle leaving the well for the box. The peak sits roughly where the well and the box carry equal Boltzmann weight. The question is who leaves first.

- **Quantum:** a single state at $-\hbar^2\alpha^2/2m$ (half the depth), with zero entropy. It leaves when $e^{a/2}\approx L/\lambda_{\rm th}$.
- **Classical:** it sits deeper, at the floor $-\hbar^2\alpha^2/m$, but pays an entropy cost to be localized in the bottom of the well. Its weight is $e^a\,k_BT/\hbar\omega_0$, with $\hbar\omega_0=\sqrt{2}\,\hbar^2\alpha^2/m$. It leaves when this equals $L/\lambda_{\rm th}$.

The zero-point energy weakens the quantum binding more than localization weakens the classical one, so *the quantum particle desorbs at a lower temperature* ($\alpha L=1000$: estimates 0.21 vs 0.27, computed peaks 0.185 vs 0.252, in units of $\hbar^2\alpha^2/2m$). Just below the quantum peak, the quantum particle is already absorbing its desorption heat while the classical one is still bound, with $C_v^{\rm cl}\approx k_B$. That is the window where $C_v^{\rm q}>C_v^{\rm cl}$. A bigger box gives more states to desorb into and a sharper transition, so the excess grows with αL: $\max(C_v^{\rm q}-C_v^{\rm cl}) = +0.34k_B$, $+2.59k_B$ and $+4.80k_B$ at $k_BT = 0.21$, 0.18 and $0.17\,(\hbar^2\alpha^2/2m)$ for $\alpha L=100$, 400 and 1000, inside the windows $[0.18, 0.24]$, $[0.13, 0.22]$ and $[0.11, 0.20]$. It is a *shift* of the peak, not a bigger peak: right after the window comes a deficit ($-9.1k_B$ at $\alpha L=1000$). **The box alone** (no well) also has a small excess, $C_v^{\rm q}-C_v^{\rm cl} = k_B\,r(1-r)/2(2-r)^2$ with $r=\lambda_{\rm th}/L$, at most $k_B/16$ (cf. Fig. 4.3 of Ref. [11]). Inside the windows above it is at most $\approx 0.01k_B$, so the anomaly comes from the well.

6 Homogeneous strokes: a heuristic

Ref. [1] lists both pieces of Eq. (1) as scale-invariant: the well under $\alpha\to\alpha/\gamma$ and the box under $L\to\gamma L$. Driving both with the *same* γ (so αL is fixed) gives $V(x)\to\gamma^{-2}V(x/\gamma)$: the same shape, stretched by γ in length and scaled by γ^{-2} in energy (driving α or L alone changes αL, i.e. the shape, and breaks what follows). The kinetic energy $\hbar^2k^2/2m$ scales the same way, since $k\to k/\gamma$. Hence $E_n\to E_n/\gamma^2$ for *every* n, which is homogeneous level scaling, $E_n^c = q\,E_n^h$ with $q=\gamma^{-2}$, in the sense of Refs. [12, 11]. In words, the stroke is only a change of units, and since the physics depends only on αL and $k_BT/(\hbar^2\alpha^2/2m)$, it changes nothing but the temperature scale:

- $Z(\beta)\to Z(\beta/\gamma^2)$ for Z_q and for $Z_{\rm cl}$ (substitute $x=\gamma y$), so $C_v(T)\to C_v(\gamma^2T)$. The stroke slides the whole curve along T, anomaly window included, without changing its shape or height.
- The working medium stays thermal, so the heat and work of the Otto cycle are integrals of C_v alone, $W=(q-1)\int_{T_c/q}^{T_h}C_v\,\mathrm{d}T$ (Eq. (4.17) of Ref. [11]; equivalently Eq. (2) of Ref. [12], whose q is our $1/q$). The efficiency is the same quantum and classical, $\eta=1-q$. If $[T_c/q,\,T_h]$ lies inside the window, the quantum engine extracts *more work at equal efficiency*.

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